2B 1 THE PERIODIC TABLE

SPECIFICATION REFERENCE
2.15 | 2.16 PART

LEARNING OBJECTIVES

- Understand that electronic configuration determines the chemical properties of an element.
- Know that the Periodic Table is divided into blocks, such as s, p, and d.

GROUPS, PERIODS AND BLOCKS

GROUPS

We call the vertical columns in the Periodic Table **groups**. All the elements in a main group (i.e. Groups 1 to 8 inclusive) contain the same outer electronic configuration. In **table A**, *n* represents the quantum shell. For lithium to neon (Li to Ne), n = 2; for sodium to argon (Na to Ar), n = 3, etc.

	GROUP 1	GROUP 2	GROUP 3	GROUP 4
outer electronic configuration	ns^1	ns^2	ns^2np^1	ns^2np^2

	GROUP 5	GROUP 6	GROUP 7	GROUP 8
outer electronic configuration	ns^2np^3	ns^2np^4	ns^2np^5	ns^2np^6

table A Outer electronic configuration of Groups 1–8 in the Periodic Table.

The elements within a group have similar chemical properties because they have the same outer electronic configuration.

PERIODS

We call the horizontal rows in the Periodic Table **periods**. All elements in a period have the same number of quantum shells containing electrons. For example, all elements in Period 2, lithium to neon (Li to Ne), have electrons in both the first and second quantum shells.

BLOCKS

The Periodic Table is also divided into blocks.

The s-block consists of the elements in Groups 1 and 2. An s-block element has its highest energy electron in an s orbital.

DID YOU KNOW?

We cannot define the d-block in the same way as the s- and p-blocks, because the electrons in the d orbitals are of lower energy than the electrons in the outer s orbital. For example, 3d has a lower energy than 4s for the d-block elements in the first row.

The p-block consists of the elements in Groups 3 to 8 inclusive. A p-block element has its highest energy electron in a p-orbital.

The d-block consists of the elements scandium to zinc (Sc to Zn) in Period 4 and yttrium to cadmium (Y to Cd) in Period 5. The number of electrons in the d orbitals gradually increases from left to right across the table. Therefore, a d-block element can be defined as one in which the d sub-shell is being filled.

CHECKPOINT

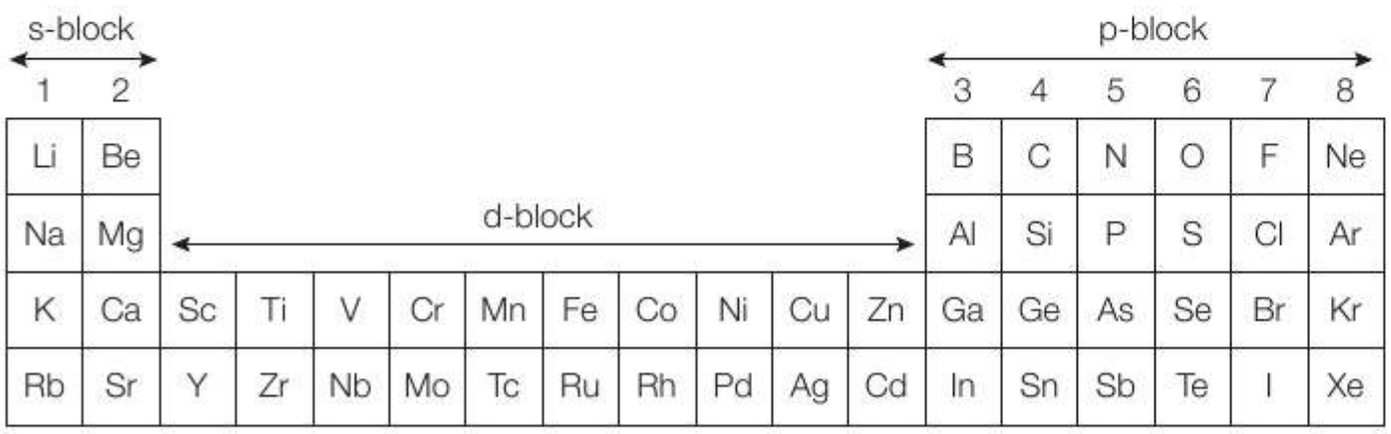
1. What is meant by the terms
   (a) s-block element and
   (b) p-block element?

2. Explain why the chemical properties of the elements in Group 1 are very similar.

3. The outer electronic configuration of carbon is $2s^22p^2$.
   Work out the outer electronic configuration of arsenic (As). Explain how you arrived at your answer.

SUBJECT VOCABULARY

groups the vertical columns in the Periodic Table
periods the horizontal rows in the Periodic Table



▲ **fig A** The s-, d- and p-blocks in the Periodic Table.

2B 2 PERIODIC PROPERTIES

SPECIFICATION REFERENCE
2.17 2.18 PART

LEARNING OBJECTIVES

Be able to explain:

- the trends in melting and boiling temperatures of the elements of Periods 2 and 3 of the Periodic Table in terms of the structure of the element and the bonding between its atoms or molecules;
- the specific trends in ionisation energy of the elements across Periods 2 and 3 of the Periodic Table.

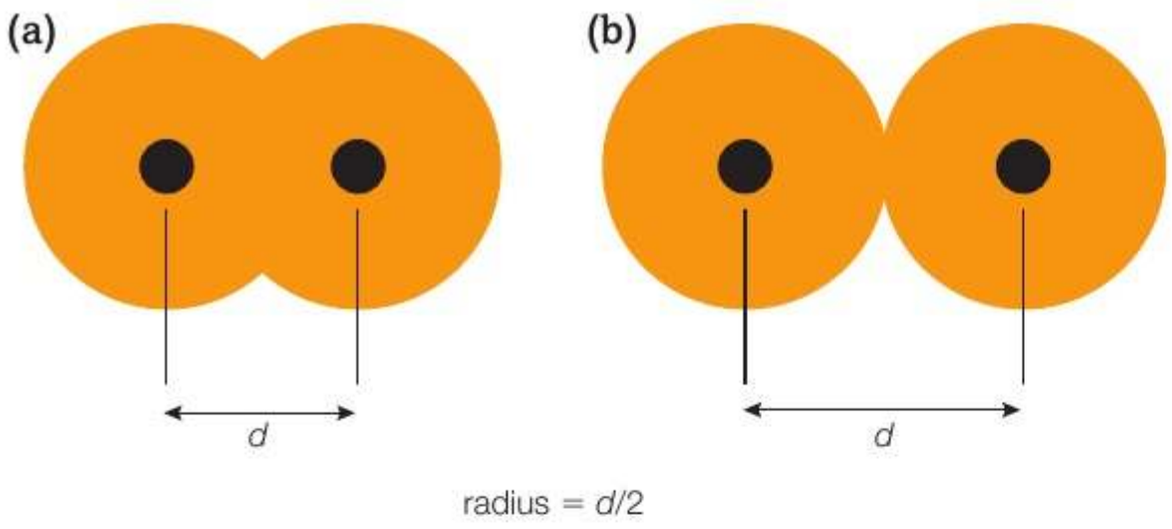
WHAT ARE PERIODIC PROPERTIES?

The elements in a period exhibit **periodic properties** (also sometimes called **periodicity**).

We can illustrate the idea of periodic properties by looking at the elements in Periods 2 and 3. We have already seen one example of periodic properties in Periods 2 and 3 – the regular repeating pattern of electronic configurations from ns^1 through to ns^2np^6. Other examples are the trends in atomic radii, melting and boiling temperatures, and first ionisation energies.

ATOMIC RADII

The atomic radius of an element is a measurement of the size of its atoms. It is the distance from the centre of the nucleus to the boundary of the electron cloud. Since the atom does not have a well-defined boundary, we can find the atomic radius by determining the distance between the two nuclei and dividing it by two.



▲ **fig A** Measuring (a) the covalent radius and (b) the van der Waals radius.

Diagram (a) in **fig A** shows two bonded atoms. The atoms are closer together than they are in diagram (b) (where they are only just touching).

In diagram (a), we are measuring the covalent radius.

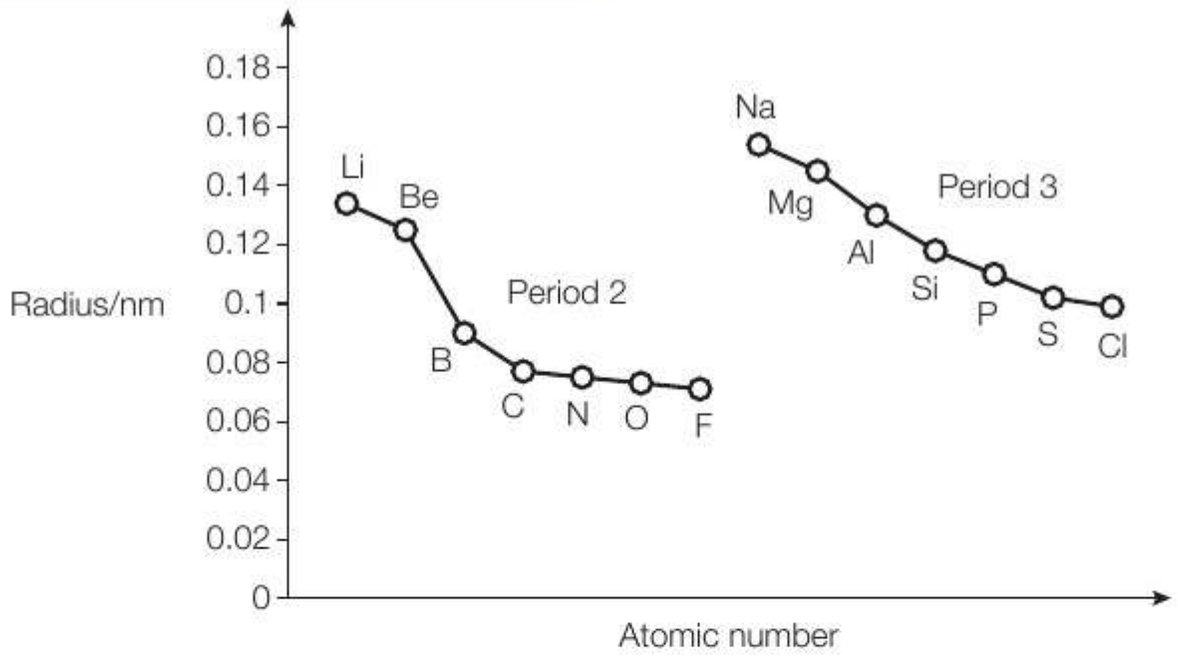
The radius we are measuring in diagram (b) is called the van der Waals radius. This is the only radius that we can determine for neon and argon, because they do not bond with other elements.

There is a third radius that is used for metals. It is called the metallic radius.

Fig B shows the trend in covalent radii across the second period (lithium to fluorine, or Li to F) and third period (sodium to chlorine, or Na to Cl) measured in nanometres (10^{-9} m).

DID YOU KNOW?

Different types of radii give different measurements for the same element. The van der Waals radius is always larger. So always compare like with like when you are looking at the trends in atomic radii.



▲ **fig B** Trend in covalent radii across the second and third periods.

You can see that the radius decreases across each period. This is because as the number of protons in the nucleus increases, so does the nuclear charge. This results in an increase in the attractive force between the nucleus and the outer electrons. This increase in attractive force offsets (counterbalances) the increase in electron–electron repulsion as the number of electrons in the outer quantum shell increases.

MELTING AND BOILING TEMPERATURES

Table A below illustrates the changes in melting and boiling temperatures for the elements in Periods 2 and 3.

PERIOD 2	Li	Be	B	C (DIAMOND)	N	O	F
melting temperature / °C	181	1278	2300	3550	–210	–218	–220
boiling temperature / °C	1342	2970	3927	4827	–196	–183	–188
type of bonding	metallic	metallic	covalent	covalent	covalent	covalent	covalent
structure	giant lattice	giant lattice	giant lattice	giant lattice	simple molecular	simple molecular	simple molecular

PERIOD 3	Na	Mg	Al	Si	P	S	Cl
melting temperature / °C	98	649	660	1440	44	113	–101
boiling temperature / °C	883	1107	2467	2355	280	445	–35
type of bonding	metallic	metallic	metallic	covalent	covalent	covalent	covalent
structure	giant lattice	giant lattice	giant lattice	giant lattice	simple molecular	simple molecular	simple molecular

table A Changes in melting and boiling temperatures for Period 2 and 3 elements.

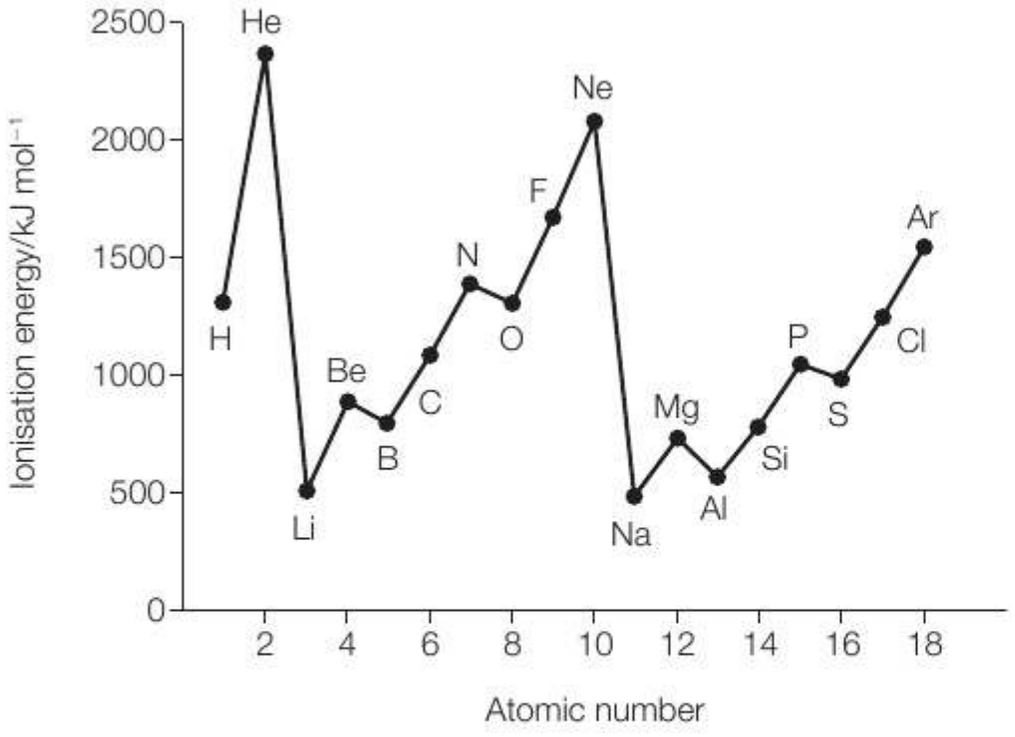
EXAM HINT

If you are asked to explain trends in melting points, you do not need to be able to remember specific melting points of elements. You only need to be able to explain the trends in terms of bond type.

You may have noticed that the elements with giant lattice structures have high melting and boiling temperatures, and those with simple molecular structures have low melting and boiling temperatures. We will explain the reason for this in **Topic 3**.

FIRST IONISATION ENERGIES

Fig C is a plot of first ionisation energy for the first three Periods.



▲ **fig C** First ionisation energy for Periods 1–3.

HYDROGEN AND HELIUM

The electronic configurations of hydrogen and helium are $1s^1$ and $1s^2$, respectively.

We can explain the increase in first ionisation energy from hydrogen to helium by the increase in nuclear charge from 1 to 2 as an extra proton is added. This increase in nuclear charge more than offsets the increase in electron–electron repulsion in the 1s orbital as a second electron is added.

LEARNING TIP

You may hear or read about a suggestion that the first ionisation energy of boron is less than that of beryllium, because by losing an electron the boron atom will acquire a full 2s orbital and so become more stable. In fact, because energy has to be supplied to the boron atom in order to remove an electron, the ion formed is energetically less stable than the atom because it has a higher energy value.

ANOMALIES

We have already explained the general increase in first ionisation energy across the period from lithium to neon (Li to Ne) and from sodium to argon (Na to Ar), in terms of the increasing nuclear charge.

However, you will notice that there are two anomalies in each case. The first ionisation energy of the Group 3 element is less than that of the Group 2 element, and the first ionisation energy of the Group 6 element is less than that of the Group 5 element.

First, consider the case of beryllium (Be) and boron (B).

The electronic configurations are:

Be: $1s^2\ 2s^2$

B: $1s^2\ 2s^2\ 2p^1$

Although the nuclear charge of the boron atom is greater than that of the beryllium atom, the outer electron of boron has more energy, since it is in a 2p orbital as opposed to the 2s orbital for beryllium. For this reason, the energy required to remove 2p electron in boron is less than the energy required to remove a 2s electron from a beryllium atom. In addition, the 2p electron in boron experiences greater electron–electron repulsion (i.e. greater shielding) because there are two inner electron sub-shells as opposed to only one in the beryllium atom.

A similar argument applies to magnesium and aluminium (Mg and Al), except that in this case it involves the 3s and 3p electrons.

LEARNING TIP

You may hear or read about an explanation stating that the outer electron in the boron atom is further from the nucleus than the outer electron in the beryllium atom. This is not correct. The boron atom is smaller than the beryllium atom as shown in **fig B**.

NITROGEN AND OXYGEN

Now consider nitrogen (N) and oxygen (O).

The electronic configurations are:

N: $1s^2\ 2s^2\ 2p_x^1 2p_y^1 2p_z^1$

O: $1s^2\ 2s^2\ 2p_x^2 2p_y^1 2p_z^1$

The first electron removed from the oxygen atom is one of the two paired electrons in the $2p_x$ orbital.

The presence of two electrons in a single orbital increases the electron–electron repulsion in this orbital. So less energy is required to remove one of these electrons than is required to remove a 2p electron from a nitrogen atom, despite the larger nuclear charge of the oxygen atom.

LEARNING TIP

You may see an explanation stating that the first ionisation energy of oxygen is less than that of nitrogen, saying this is because when the oxygen atom loses an electron, it acquires a stable half-full 2p sub-shell. However, there is no special stability associated with a half-full sub-shell. Also, as with the boron atom, energy must be supplied to the oxygen atom in order to remove an electron, so the ion formed is energetically less stable than the atom.

You may also hear or read about an explanation stating that the outer electron in the oxygen atom is further from the nucleus than the outer electron in the nitrogen atom. This is not correct. The oxygen atom is smaller than the nitrogen atom as shown in **fig B**.

CHECKPOINT

SKILLS REASONING

1. An element has very high melting and boiling temperatures.
   (a) What type of structure is it likely to have?
   (b) Which physical property would help you determine whether the bonding was metallic or covalent? Explain your answer.
2. Explain why the first ionisation energy of helium is higher than that of hydrogen.
3. Explain why the first ionisation energy of lithium is much lower than that of helium, even though the lithium atom has a greater nuclear charge.
4. Would you expect the first ionisation energy of gallium (Ga) to be higher or lower than that of calcium (Ca)? Explain your answer.
5. Why does neon have the highest first ionisation energy of all the elements in Period 2?

SUBJECT VOCABULARY

periodic properties (periodicity) regularly repeating patterns of atomic, physical and chemical properties, which can be predicted using the Periodic Table and explained using the electron configurations of the elements